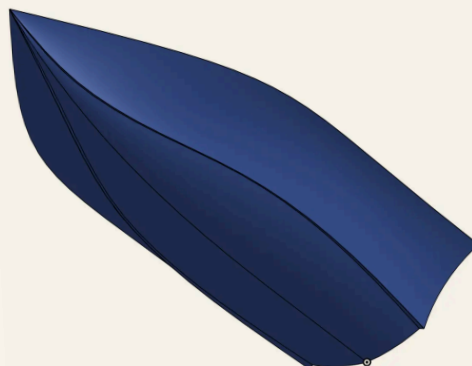


Haav

Highly Amphibious / Autonomous Vehicle



A student team building an autonomous boat from the ground up.



First-principles mixed-integer linear programming

We build the model from physics up. Each constraint cuts the space of possible boats, and the optimizer finds the fastest one left. Hull resistance comes from **Savitsky's planing-hull model**. As we build and test, we keep updating the model and adding constraints: a hybrid of simulation and empirical data.

Engineers from across UW–Madison

Each segment of the boat is covered, and members have worked at Tesla, Northrop Grumman, Xcel Energy and Milwaukee Tool.

Industrial

optimization, composites and manufacturing

Mechanical

structures, CAD and fabrication

Electrical

power, battery and high voltage

Computer

embedded, firmware and software

Our approach

Hull

Plane well and stay light.

Hybrid planing composite hull
Shape tuned by optimization
Resistance checked in CFD first

Electric

Every watt-hour to speed.

High-voltage battery storage
Propulsion sized by the model
Propulsion and cooling co-designed

Autonomous

Manage drift and controls.

GPS, IMU and LIDAR sensing
Firmware protects the battery
Software plans the route

Timeline

Oct 25

Hull design frozen

Nov 1

Preliminary design review

Nov 10

Critical and long-lead parts are ordered

Nov 20

Mechanical and electrical design frozen

Nov 20–Dec 20

Build the hull, composite parts and any custom-milled parts

Jan 8

First float and tow test

Jan 8–30

Manufacture and assemble everything

Jan 8–30

Build likely spare parts, in parallel

Jan 31

First nautical mile

Feb 1–14

Slow drives on the water

Feb 15

Start increasing speed

Until April

Testing / Tuning

Want to help build it?

We are looking for funds, parts, advisors and people to build our BOM as we design.

bit.ly/haav

Marley Barrett, Project lead
mhbarrett@wisc.edu